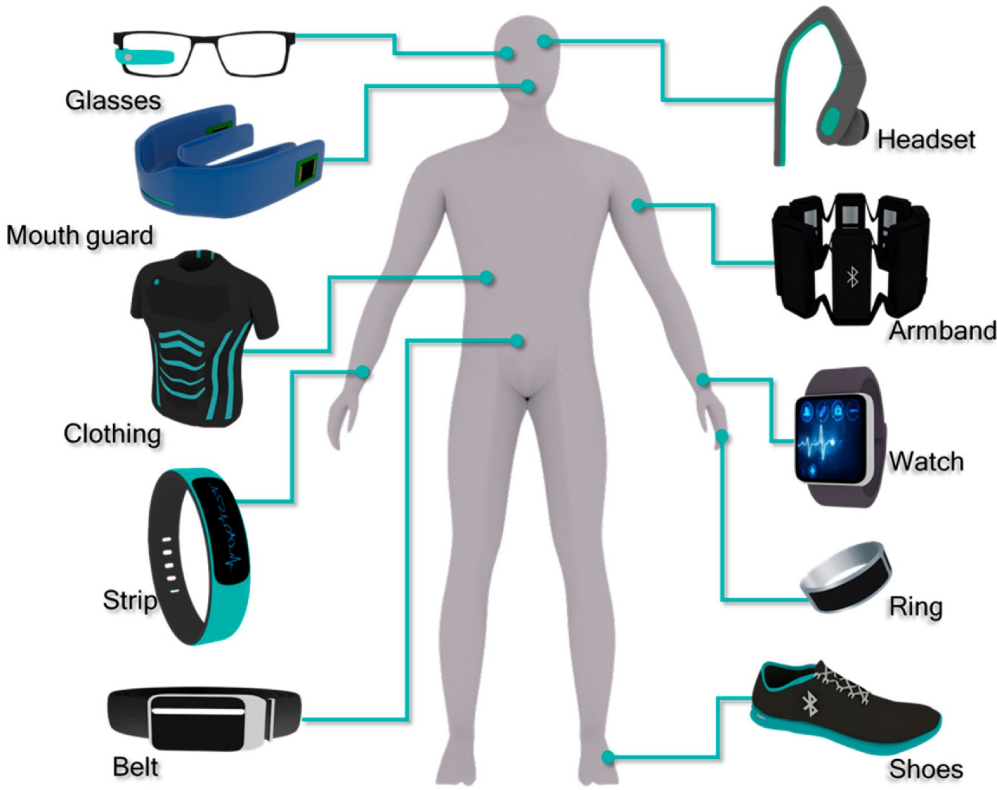
Introduction to Ubiquitous Computing:
Core Concepts and Applications, Week 1



Department of Software Engineering
Faculty of Computing
UTM

Introduction

- ✔ WHAT is Ubiquitous Computing?
- ✔ Evolution of Ubiquitous Computing
- ✔ Architecture
- ✔ Computing Technologies
- ✔ Smart DEI Framework
- ✔ Requirements/properties Framework
- ✔ Interaction Framework
- ✔ The application of Ubiquitous Computing



WHAT is Ubiquitous Computing?

- Also known as **pervasive computing or ambient intelligence**
- Computing devices are embedded into everyday objects and environments.
- Computing becomes seamlessly integrated into daily life.
- Technology operates in the background and is largely invisible to users.



Goal of Ubiquitous Computing

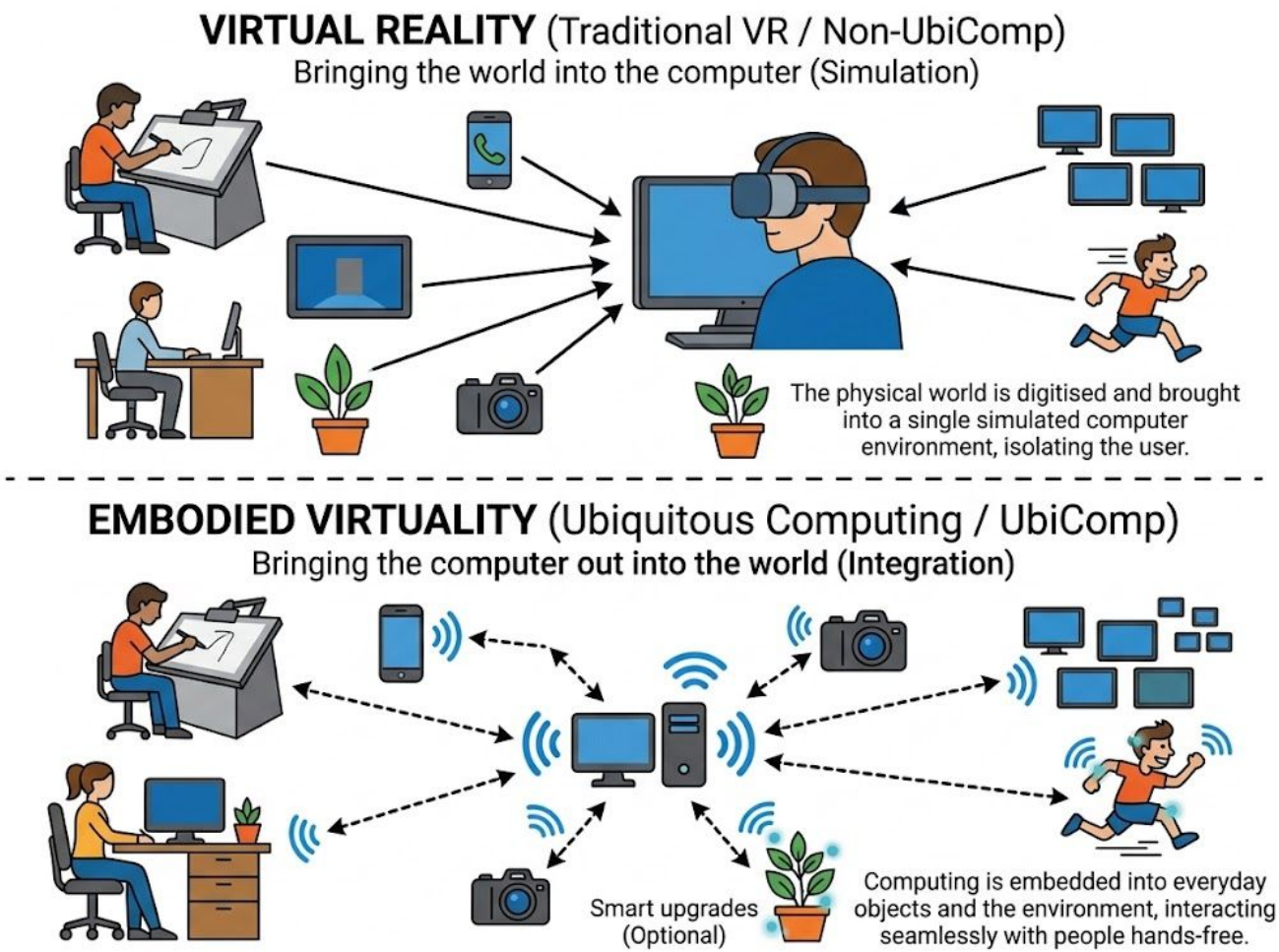
- Integrate computing capabilities seamlessly into the environment.
- Enable continuous and unobtrusive user interaction.
- Enhance convenience without requiring explicit user attention.

Vision of Ubiquitous Computing

- Provide computer-based services **anytime** and **anywhere**.
- Support natural and intuitive human interaction.
- Make technology invisible while remaining readily available.

WHAT is Ubiquitous Computing?

Ubiquitous Computing	Non – Ubiquitous Computing
Force computer (computing power) to live out in the world with people	Put people inside a computer “world”
Very difficult integration of a lot of factor such as human factors, computer science, social science and engineering	Primarily depends on computing power only



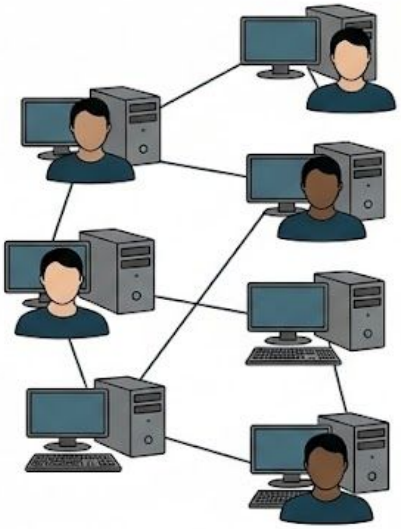
Evolution of Ubiquitous Computing?

1st Era: Mainframe Era



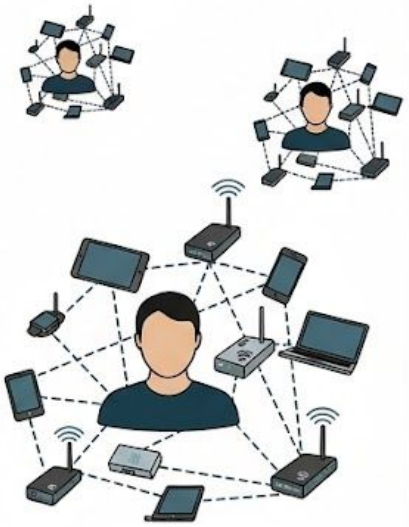
Many People to One Computer

2nd Era: PC Era



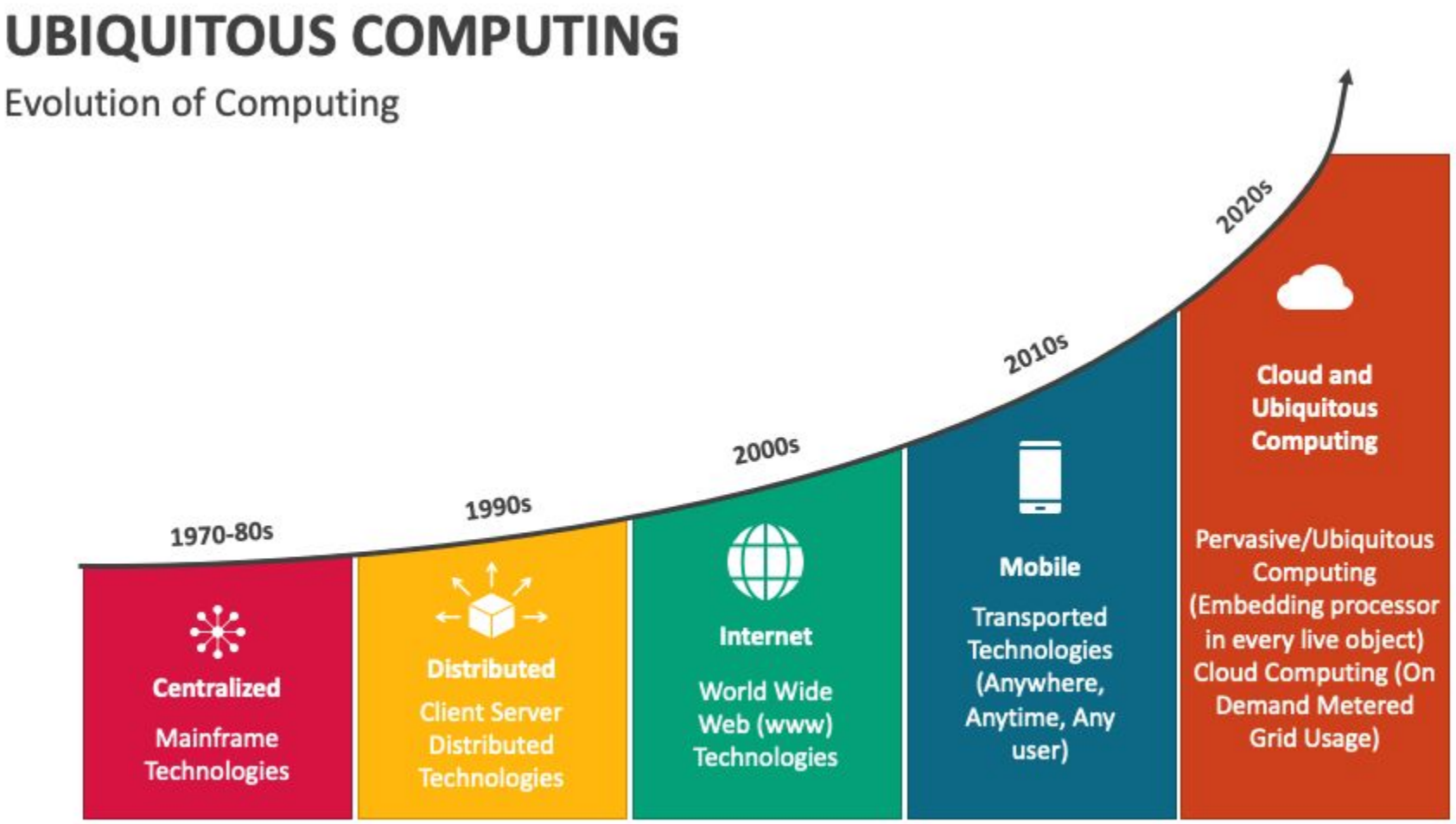
One Person to One Computer

3rd Era: Ubiquitous Computing Era



One Person using Many Computers (computing devices)

Evolution of Ubiquitous Computing?

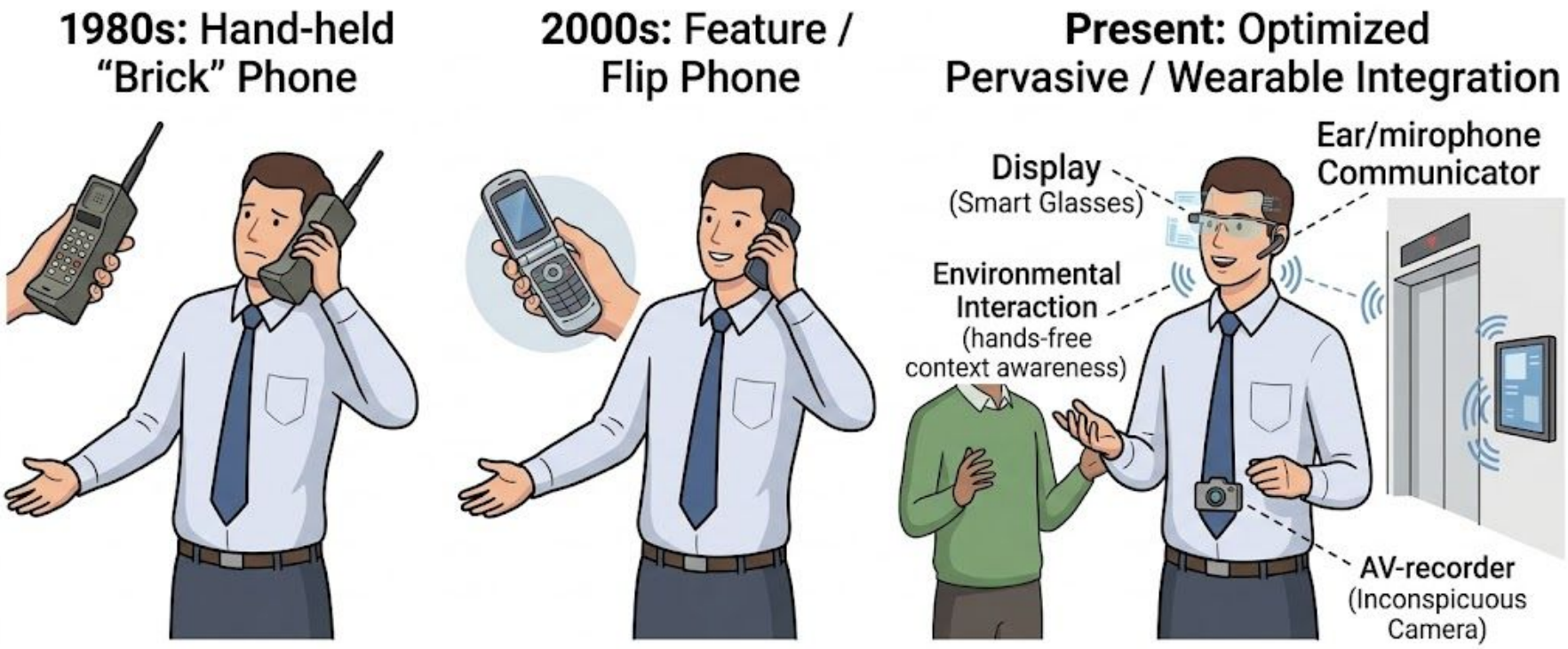


Evolution of Ubiquitous Computing?

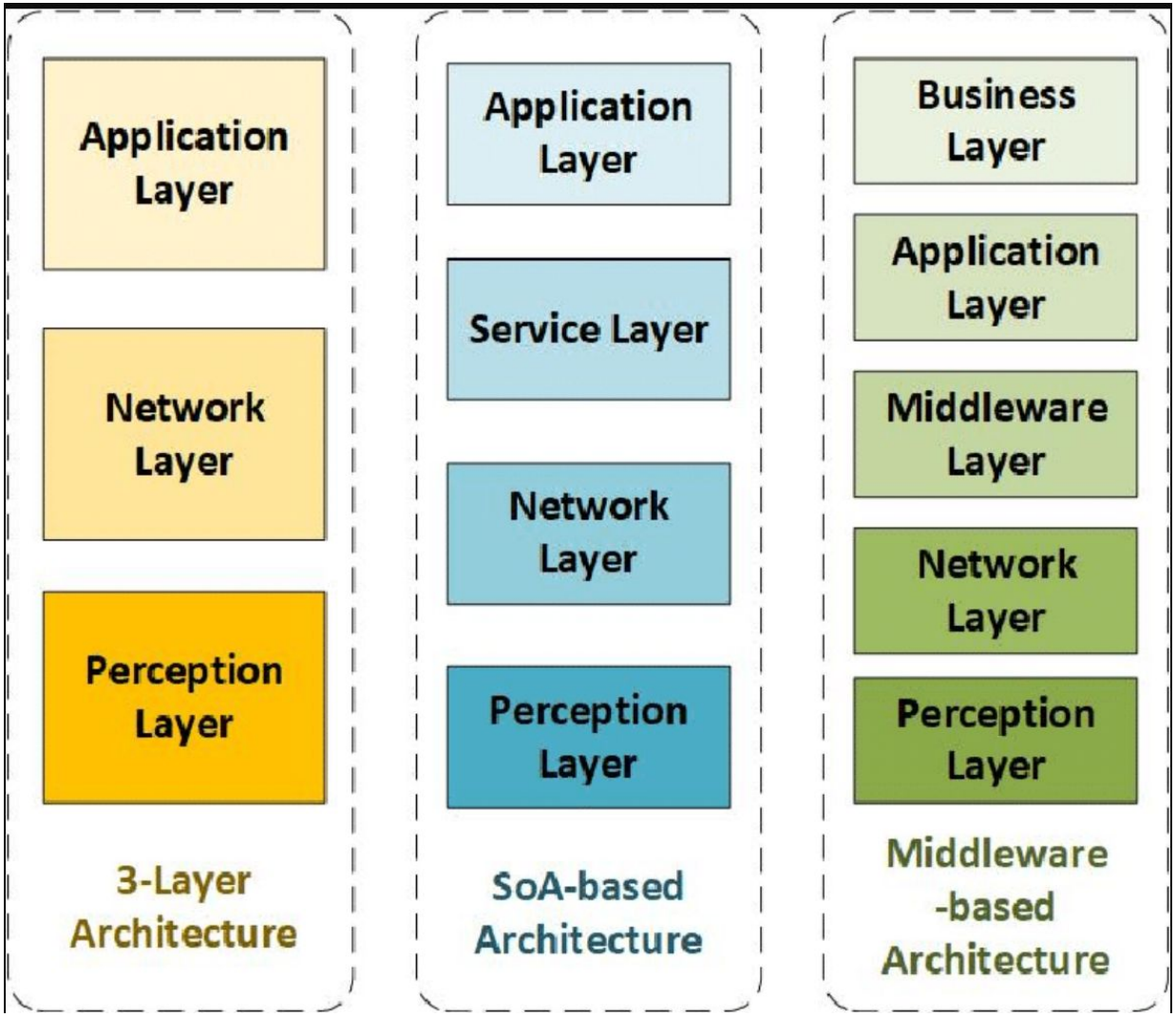
- 3 main properties for Ubiquitous Computing Systems were proposed by Weiser (1991)

1. Computers need to be networked, distributed and transparently accessible
2. Computer Interaction with Humans needs to be more hidden
3. Computers need to be aware of environment context

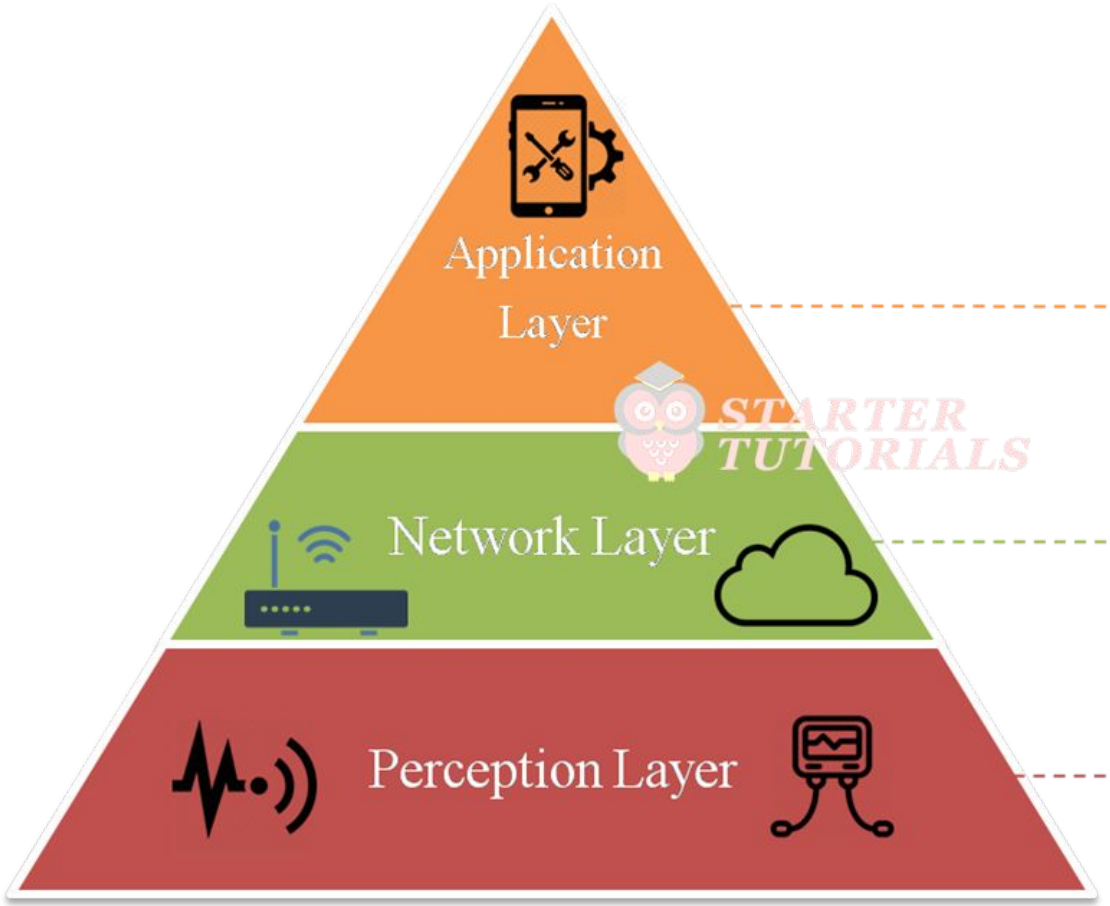
- To optimize their operation in their physical & human environment.



Most Common Architecture



3 Layer Architecture

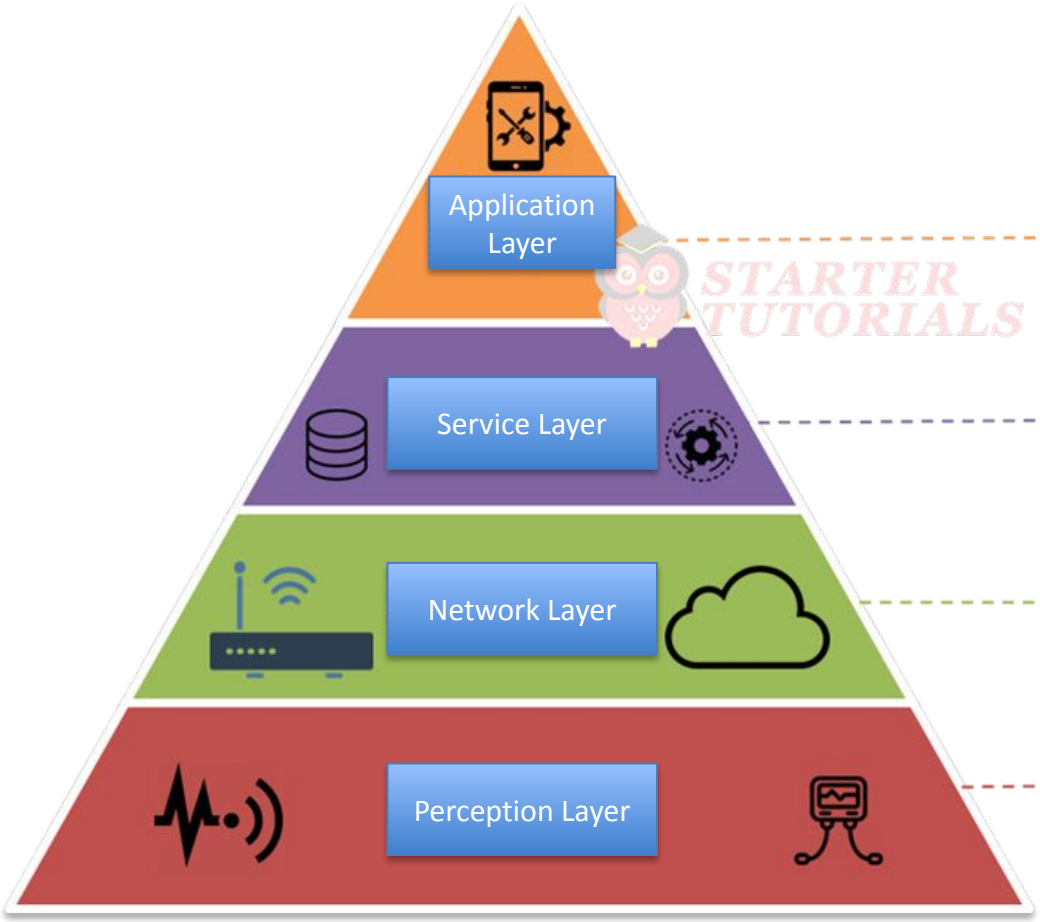


Responsible for delivering application services to the users via user interfaces.

Responsible for connecting to other devices or servers. Also includes transmitting and processing the data from sensors.

Is the physical layer which contains sensors to gather data.

SOA-based Architecture



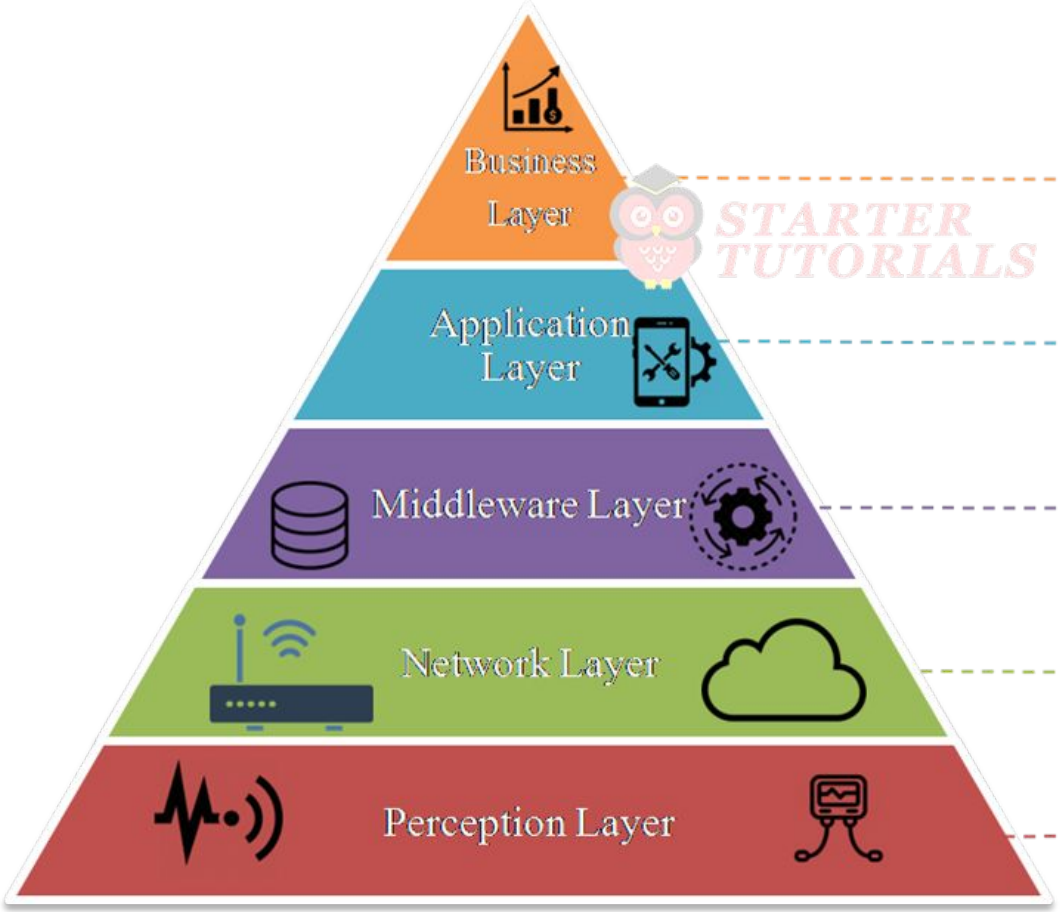
Provides interfaces to interact with the services provided by the service layer

Creates and manages services required by users and applications

Transfers the data from sensing layer to service layer.

Is the physical layer which contains sensors to gather data.

Middleware-based Architecture



Manages the whole IoT system, including applications, profit models, etc

Responsible for delivering application services to the users via user interfaces.

Stores, analyzes, and processes data coming from the network layer.

Transfers the data from perception layer to middleware layer.

Is the physical layer which contains sensors to gather data.

EVOLUTION OF COMPUTING TECHNOLOGIES

1 EMBEDDED SYSTEMS



Dedicated devices with specific functions.

Examples: Microwave, washing machine, automotive ECU

Fixed function, standalone operation

2 EMBEDDED REAL-TIME SYSTEMS



Embedded systems with strict timing constraints and high reliability.

Examples: Airbag system, pacemaker, industrial control

Deterministic response, real-time constraints

3 INTERNET OF THINGS (IoT)



Devices connected via networks that collect and exchange data.

Examples: Smart home devices, connected sensors, smart meters

Connectivity, data exchange, remote monitoring & control

4 UBIQUITOUS COMPUTING



Computing is embedded seamlessly into everyday environments.

Examples: Smart homes, smart hospitals, ambient intelligence

Context-aware, seamless, anytime-anywhere interaction

5 MOBILE COMPUTING



Computing on the move using portable devices with connectivity.

Examples: Smartphones, tablets, mobile apps

Mobility, wireless access, user-centric computing

6 WEARABLE COMPUTING



Computing integrated into wearable devices close to the human body.

Examples: Smartwatches, fitness trackers, AR glasses

Personalized, continuous sensing, body-worn

7 EDGE COMPUTING



Data processing occurs near the source (edge) instead of relying only on the cloud.

Examples: Edge gateways, smart cameras, industrial edge

Low latency, local processing, less bandwidth dependence

8 INTELLIGENT EDGE DEVICES



Edge devices with AI capabilities to perceive, learn, decide and act autonomously.

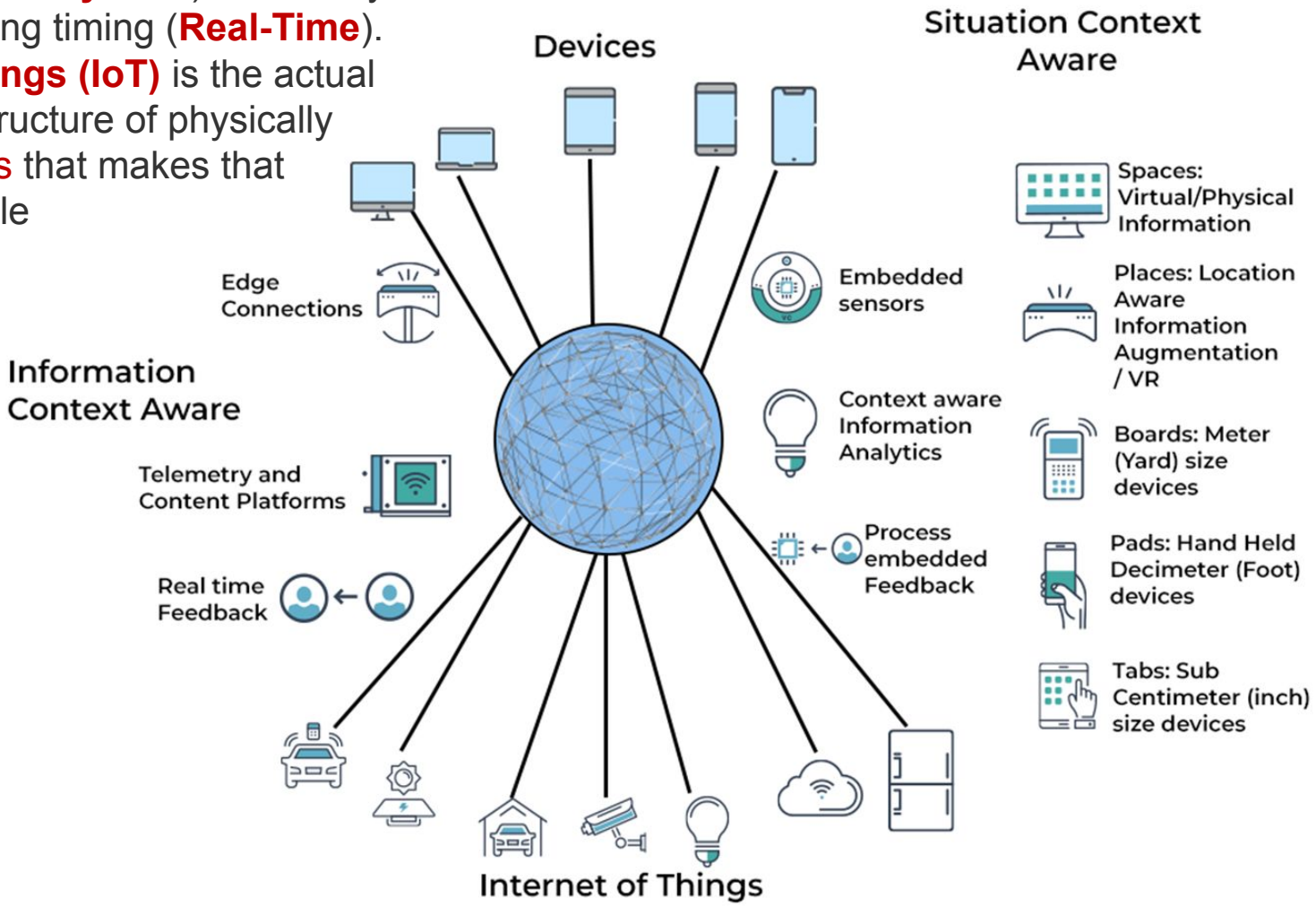
Examples: AI cameras, smart wearables, autonomous robots

AI-powered, autonomous decision-making at the edge

From isolated, dedicated systems → to connected, context-aware, intelligent edge devices that sense, communicate, process and make decisions autonomously.

Computing Technologies: UbiComp, ERT System and IoT Technologies

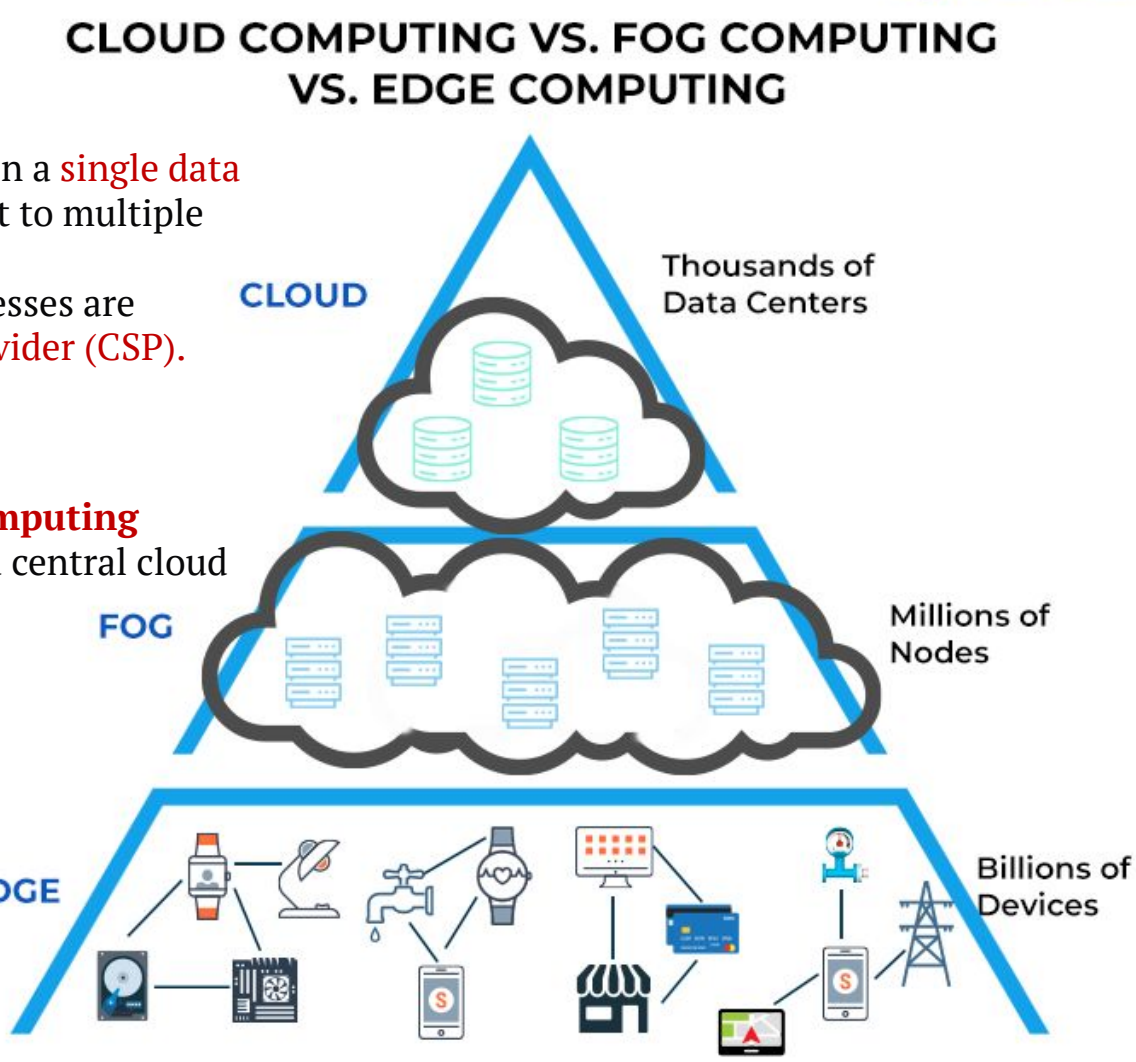
- Ubiquitous Computing (UbiComp) is the overarching philosophy of computing being available everywhere.
- Everywhere require dedicated embedded devices (**Embedded System**) with many constraints including timing (**Real-Time**).
- **or Internet of Things (IoT)** is the actual networking infrastructure of physically connected devices that makes that philosophy possible



- data and applications are converged in a single data center and delivered over the internet to multiple end point devices.
- All the resources to execute the processes are contributed by the Cloud Service Provider (CSP).

- places a **decentralized enterprise computing layer** between the source of data and a central cloud platform

- brings processing and storage systems as **close** as possible to the application, device, or component that generates and collects data.
- helps **minimize processing time** by removing the need for transferring data to a central processing system and back to the endpoint

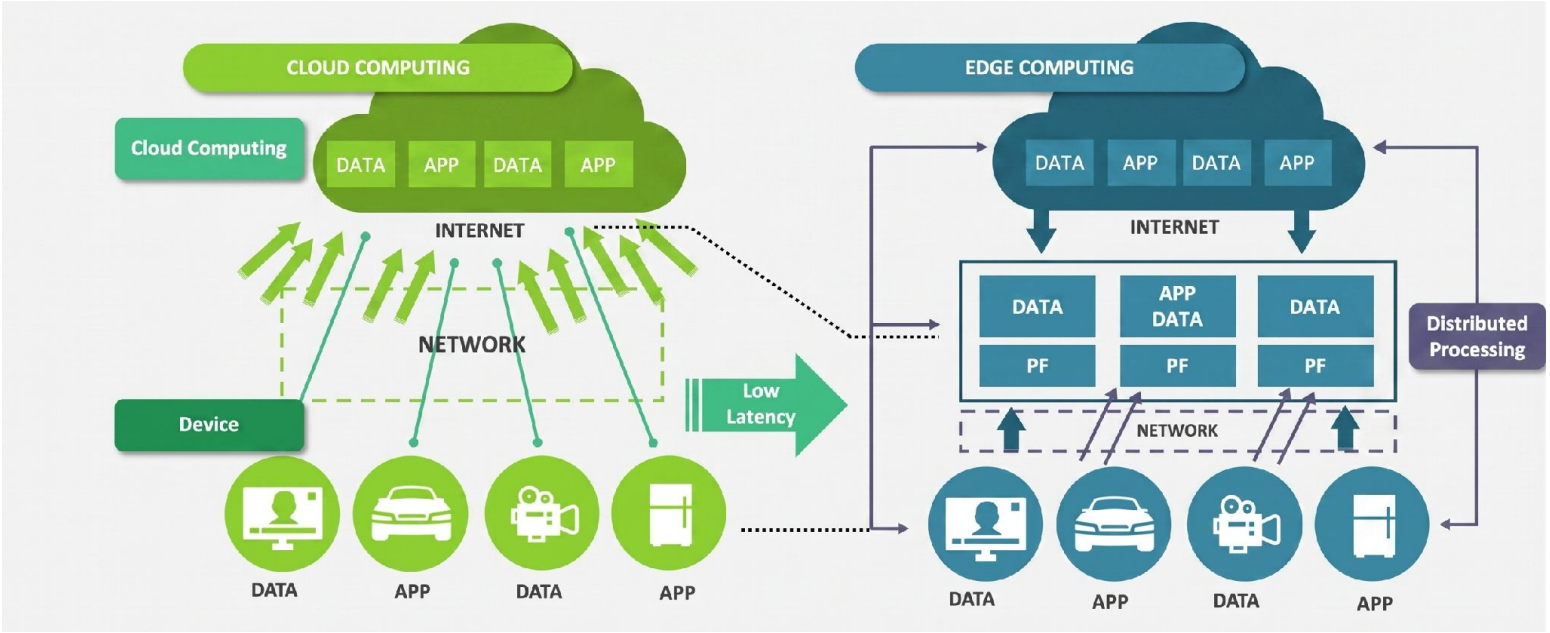


Computing Technologies

Compare	Cloud	Fog	Edge
Latency	Highest	Medium	Lowest
Scalability	High, easy to scale	Scalable within network	Hard to scale
Distance	Far from the edge	Network close to the edge	At the edge
Data analysis	Less time-sensitive data processing, permanent storage	Real-time, decides to process locally or send to cloud	Real-time, instant decision making
Computing power	High	Limited	Limited
Interoperability	High	High	Low

Computing Technologies: Edge Computing

- Devices move from just sending data to the cloud to thinking and making decisions locally at the source.

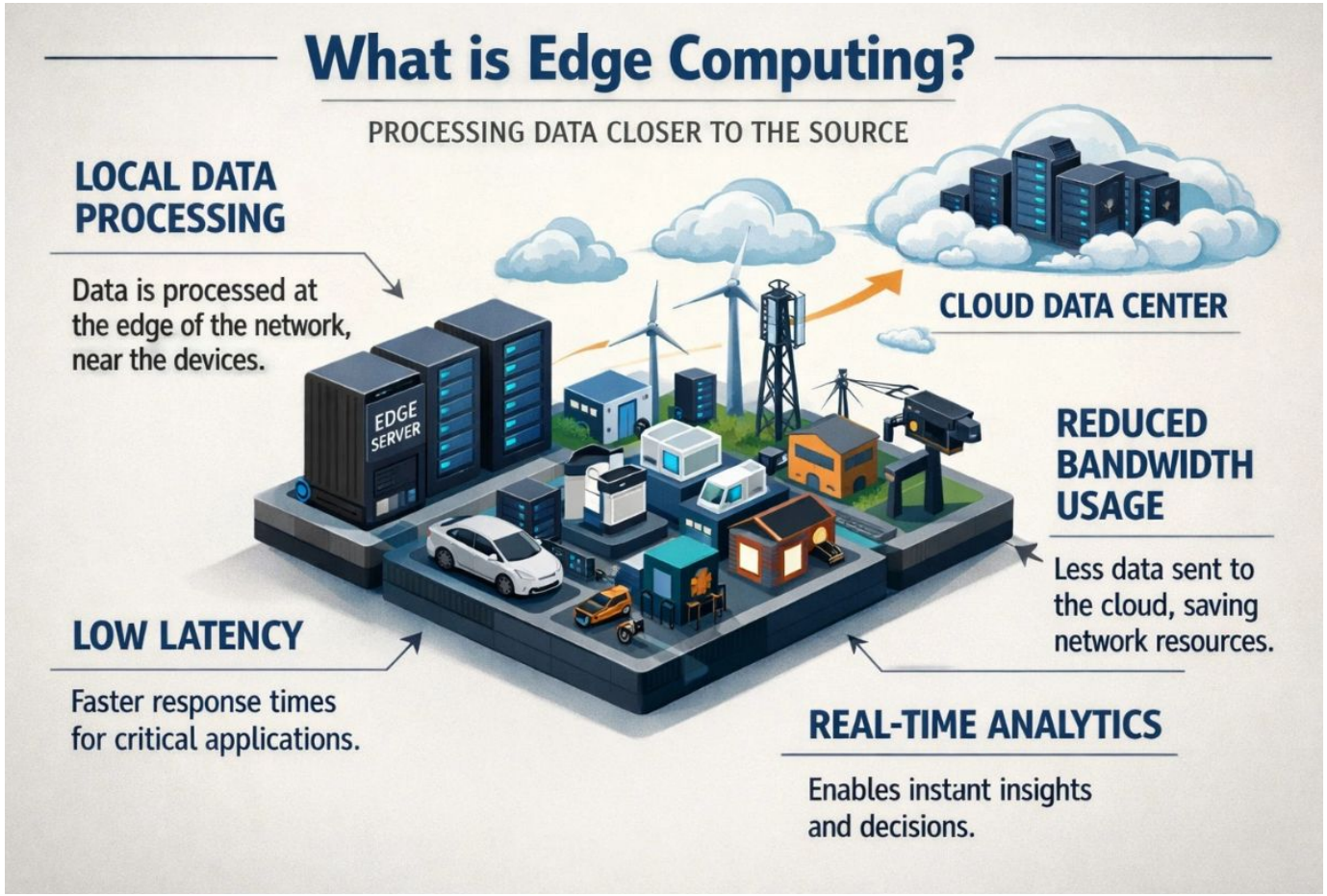


IoT Smart Connected



Intelligent Edge

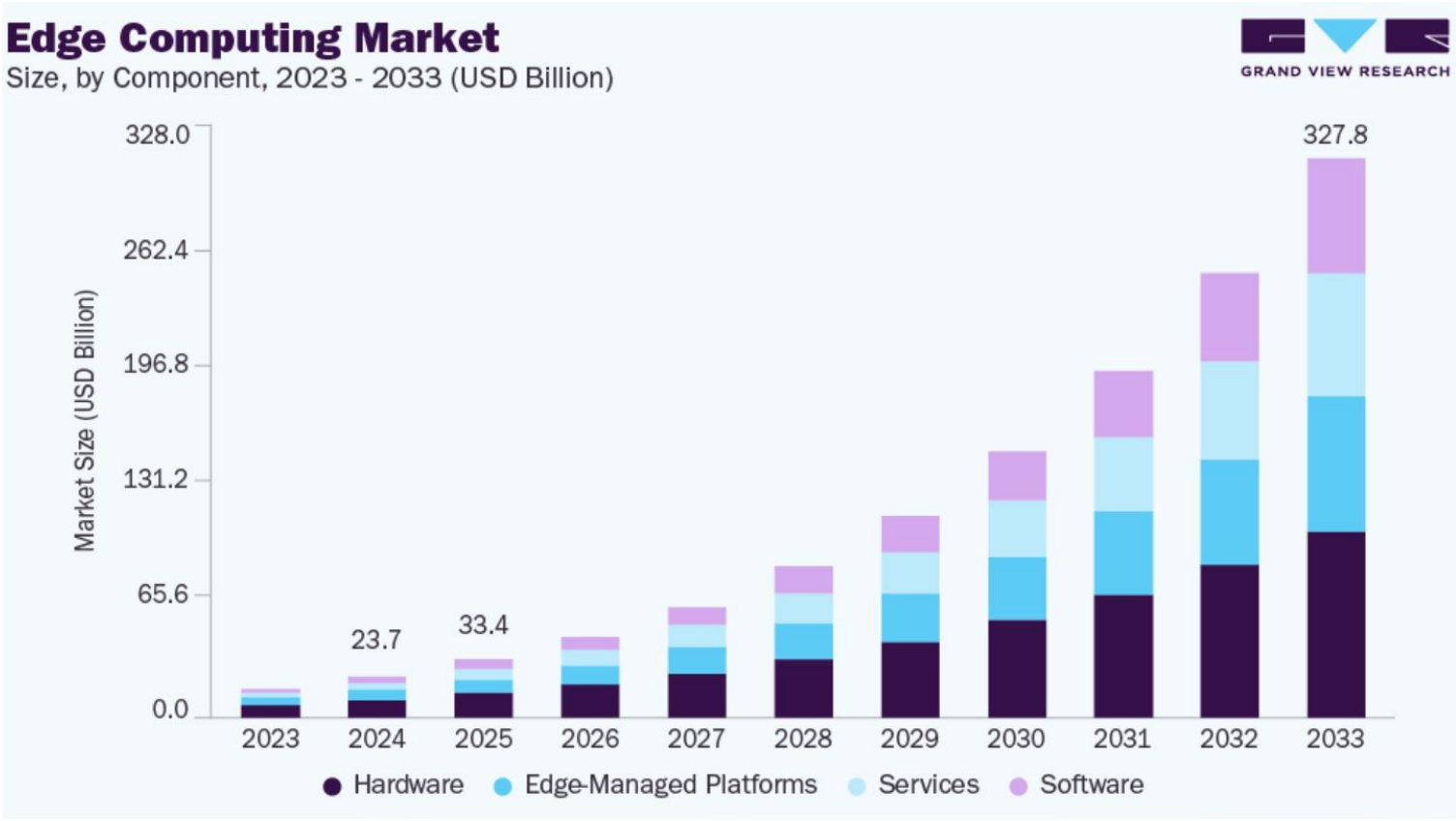
Computing Technologies: Edge Computing



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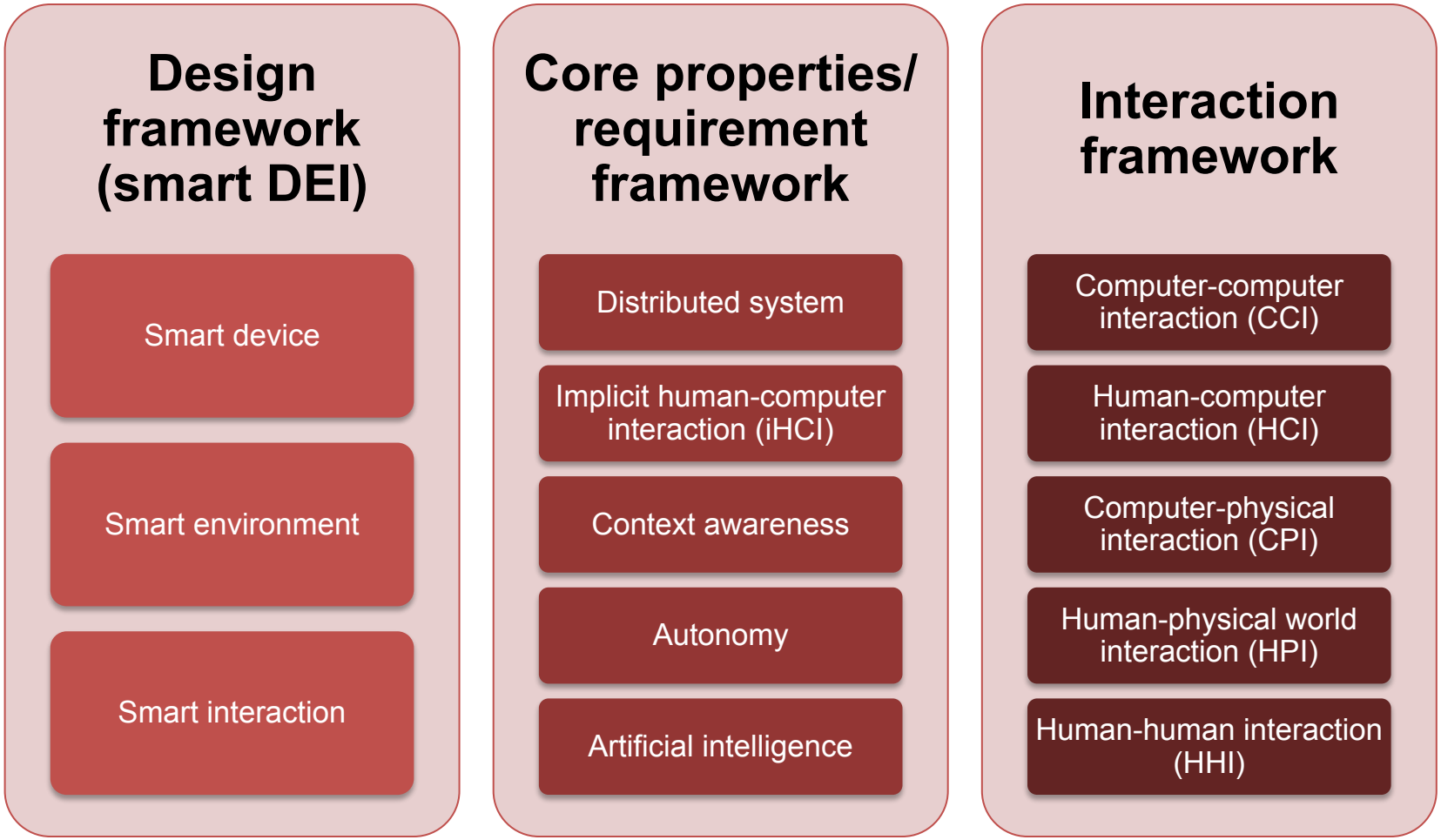
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Computing Technologies: Edge Computing



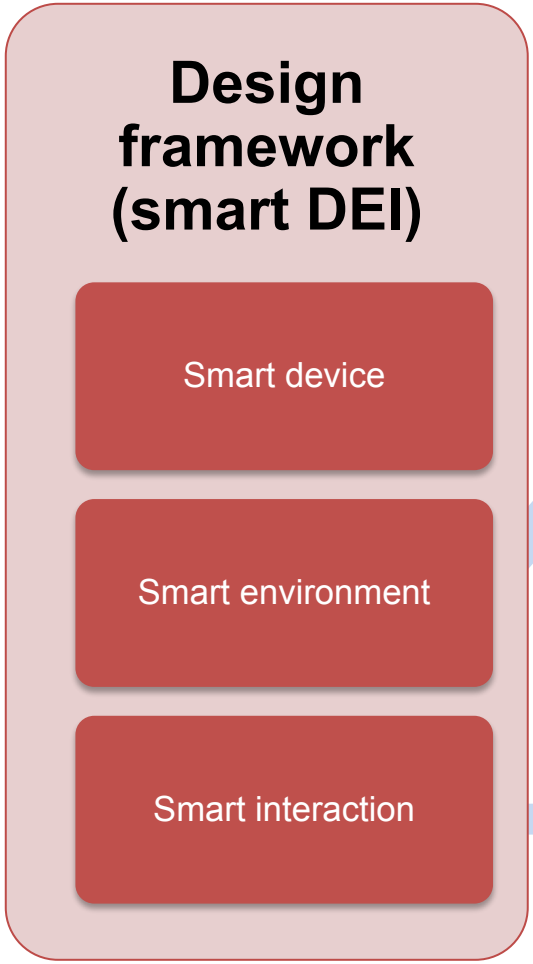
Edge Computing Market (2025 - 2033)Size, Share & Trends Analysis Report By Component (Hardware, Software, Services), By Application, By Organization Size, By Industry Vertical, By Region, And Segment Forecasts

Framework for Ubiquitous Computing



Framework: Smart Devices, Environments and Interaction (DEI*)

- 3 main types of design for ubiquitous computing systems



1. Device

Use more complex, multifunctional, mobile, personalized and private **smart devices** to ease access and embody services rather than just to virtualize them

Environment

Use **smarter environments** to sense and react to event. Eg: people entering or leaving spaces, through mobile device interaction.

Interaction

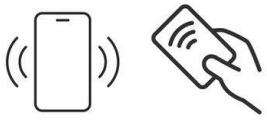
Use more service access devices with simpler functions and allow them to interoperate - **smarter interaction** between devices

* Smart DEI (pronounced smart ‘day’)

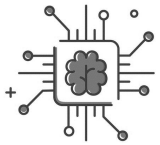
Characteristics of Smart DEI

Smart DEI Model

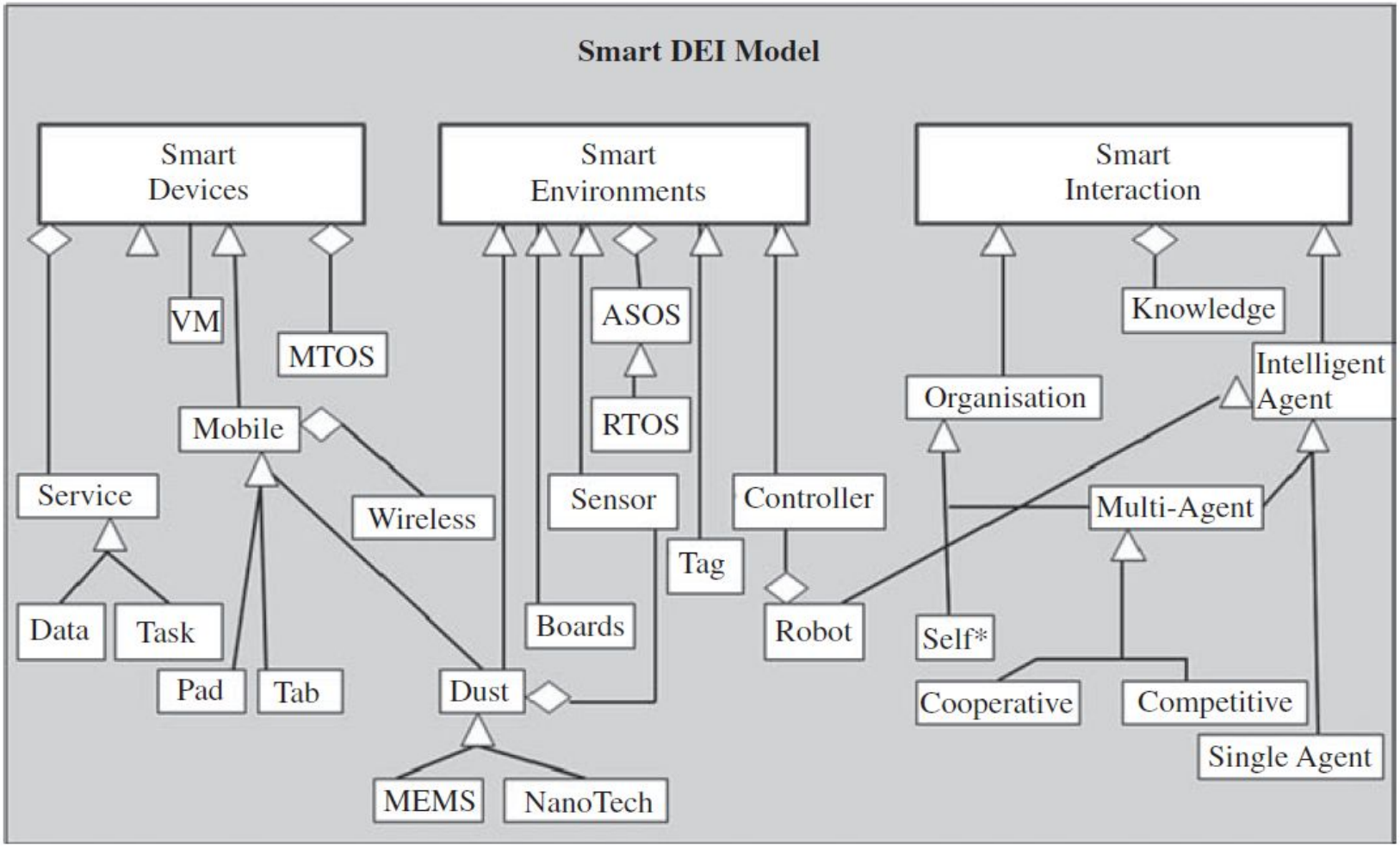
Smart Devices	Smart Environments	Smart Interactions
Examples: Smartphones, wearables.	**Examples:** Smart rooms, sensor networks, active buildings.	**Examples:** Self-regulating grids, swarm robotics.
Mobility: Highly mobile, travels with the user.	**Mobility:** Static, embedded in the physical world.	**Mobility:** Fluid, distributed across systems.
Locus of Control: Personal / User-centric.	**Locus of Control:** Environmental / Spatial.	**Locus of Control:** Software-driven / Autonomous.
Function: Multi-functional portals to volatile services.	**Function:** Single-task optimization (e.g., lighting, HVAC, tracking).	**Function:** Multi-agent collaboration without human intervention.







Framework: Smart DEI



Some of the main subtypes (triangle relationships) of smart devices, environments and interactions and some of their main aggregations (diamond relationships)

MTOS - Multi Tasking Operating System
ASOS - Application Specific OS
VM - Virtual Machine
MEMS - Micro ElectroMechanical System
RTOS - Real Time OS

Role of Mobile and Wearable Devices in Ubiquitous Computing

Mobile and wearable devices serve as the essential building blocks that bridge the digital and physical worlds, facilitating a shift from the personal computer (PC) era to an environment where one person uses many interconnected computers

Pervasive services and personal space

Smartphones act as portals to pervasive services, allowing users to access remote information and applications anywhere. They also support a personal information cloud, where a user's data, preferences and services follow them and enable personalized experiences.

Invisible and calm technology

Wearable devices integrate computers into clothing or accessories so they remain unobtrusive and comfortable. They support calm computing by working in the background while still providing constant access to information when needed.

Enable augmented and mediated reality

Head-up displays overlay digital information onto the physical world to support tasks like maintenance or surgery. They also enable mediated reality, allowing users to enhance or filter what they see and hear in real time

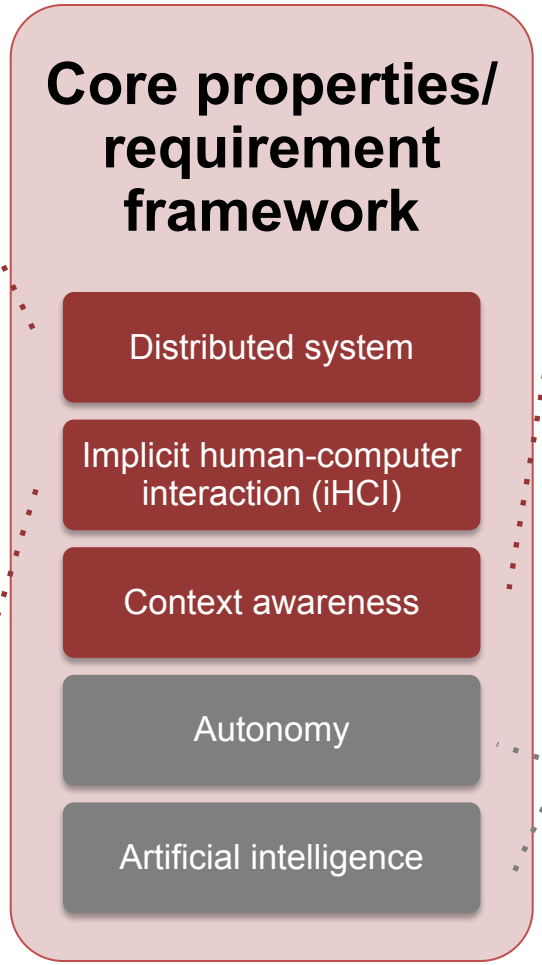
Posthuman and embedded integration

Mobile, wearable and implanted devices can extend human abilities and experiences. They also connect through local networks to interact with smart environments, creating seamless device cooperation.

Core properties / requirement

Distributed system
- Universal, seamless, heterogeneous
- Networked
- Synchronized, coordinated
- Open
- Transparent, virtual
- Mobile, nomadic

Implicit human computer/device interaction (iHCI)
- Non intrusive, hidden, invisible, calm computing
- Tangible, natural
- Anticipatory, speculative, proactive
- Affective, emotive
- User aware
- Post human
- Sense of presence immersed, virtual, mediated reality



Key properties for ubiquitous system?

Context awareness
- Sentient, unique, localized, situated
- Adaptive, active context aware
- Person aware, user aware, personalised, tailored
- Environment aware, context aware, physical context aware
- ICT awareness

Additional requirements
- Operate autonomously without human intervention
- Handle multiple dynamic interactions using intelligent decision-making to:

a) Deal with uncertain or unpredictable situations
b) Support cooperation and competition between systems
c) Enable richer interaction through shared context and goals

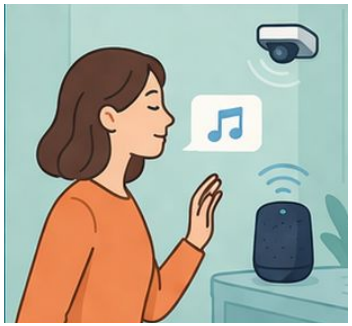
Distributed system

- Able to operate across different homogeneous environments, seamless integration of devices and environments, taking on new contexts when new resources become available
- Devices are interlinked using a network which is often wireless
- Multiple entity interaction can be coordinated synchronously or asynchronously over time and space interactions
- New components can be introduced and accessed, old ones can be modified or retired. Components can be dynamically discovered
- Systems aim to be transparent. Reduces the operational complexity of computing, acting as a single virtual system even although it is physically distributed
- Users, services, data, code and devices may be mobile



Implicit human device interaction (iHCI)

- Reduces the need for explicit user actions. It is integrated into the general ecology of the home or workplace and can be used intuitively by users
- Interaction is via natural user interfaces and physical artefact interaction that can involve gestures, touch, voice control, eye gaze control, etc.
- Improving performance and user experience through anticipated actions and user goals in relation to current context, past user context and group context. This overlaps with user context awareness
- Computing that relates to, arises from, or influences human emotions. This is also considered to be a sub type of human intelligence
- Aware of presence of user, user ID, user characteristics, current user tasks in relation to users' goals (as part of iHCI and context awareness)
- Sense of being in a world that exists outside ourselves, extending a person's normal experience across space and time (post human)
- A person is in a real time interactive environment which experiences an extended sense of presence that combines the virtual and the real, often by overlaying virtual views on real views



Context awareness

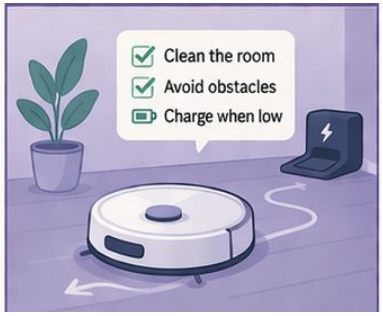
- Systems can discover and take advantage of the situation or context such as: location, time and user activity.
- Types of context awareness;
  Physical world (location, temperature, time) – Spatial & temporal awareness
  User (preferences, activity)
  Virtual (network/services status)
- Systems actively adapt to context changes in a dynamic environment rather than just present context changes to the user
- Tailored to an individual user or type of user, based on personal details or characteristics that a user provides or is gathered about a user. This may trigger system adaptation
- Sometimes, researchers use physical context awareness to refer more broadly to general context or environmental awareness.
- Helps provide relevant services only when needed.
- Can be:
  Active → system acts automatically
  Passive → system only informs user

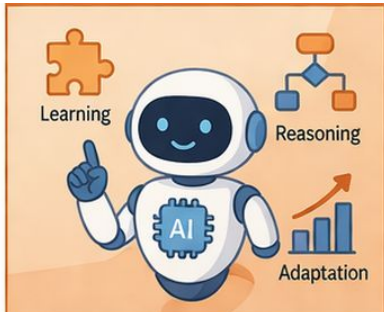


Autonomy & artificial intelligence

- Systems can operate independently without human control.
- Make decisions based on goals or policies.
- Reduce need for constant human interaction.
- Can plan, adapt, and re-plan tasks automatically.
- Improves efficiency but may reduce user control/understanding

- Systems use AI to make smarter decisions.
- Helps systems:
  - Handle uncertainty and incomplete data
  - Adapt to dynamic environments
  - Support user goals automatically
- Enables:
  - Environment modelling
  - User behaviour understanding
  - Semantic (meaning-based) interaction





3 types of system environments:

1



Human

- Human individuals and human organizations
- Human living entities are often governed by economic and social models

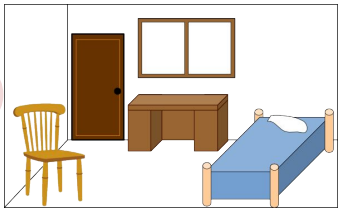
2



Computer / ICT

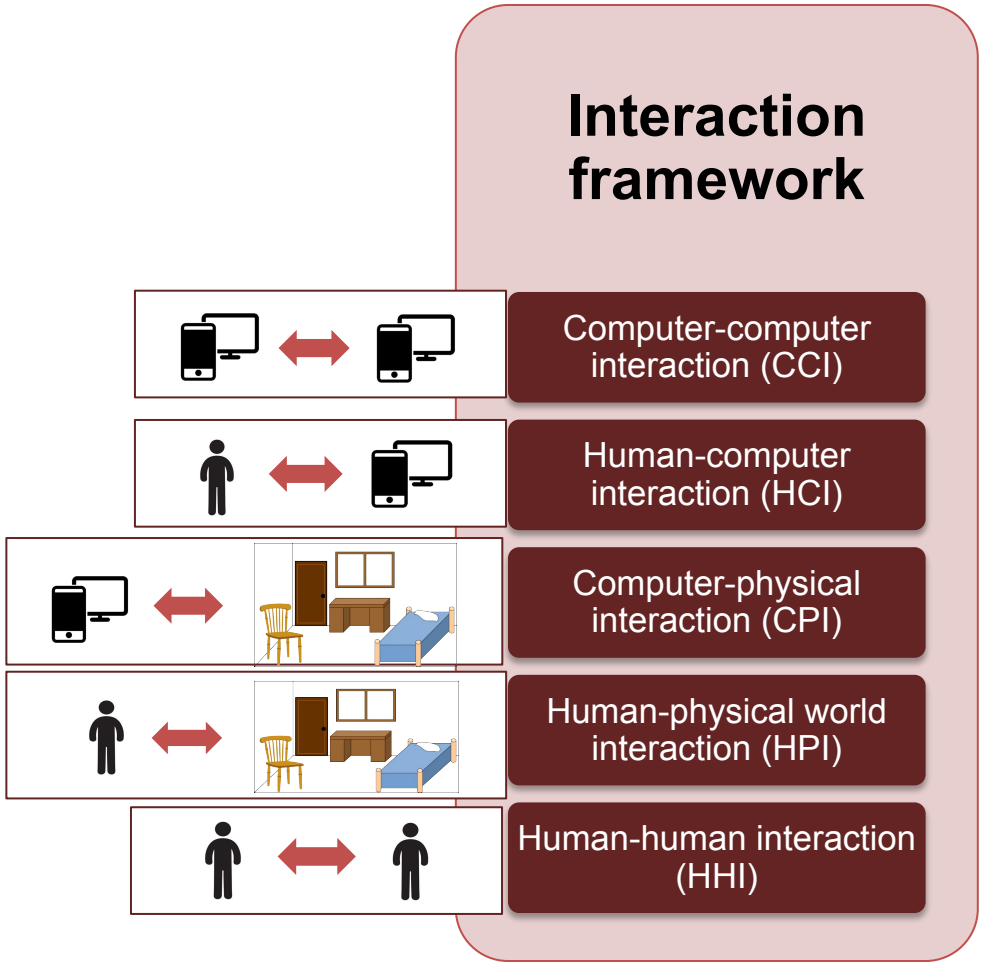
- Systems which form the ICT infrastructure, supporting services and act as middleware for that particular ICT system applications (virtual worlds)

3



Physical world / space

- Systems including ecological and biologic systems
- Physical world phenomena are governed by well known laws of physical forces such as electromagnetism and gravity



How system interact with different environment?

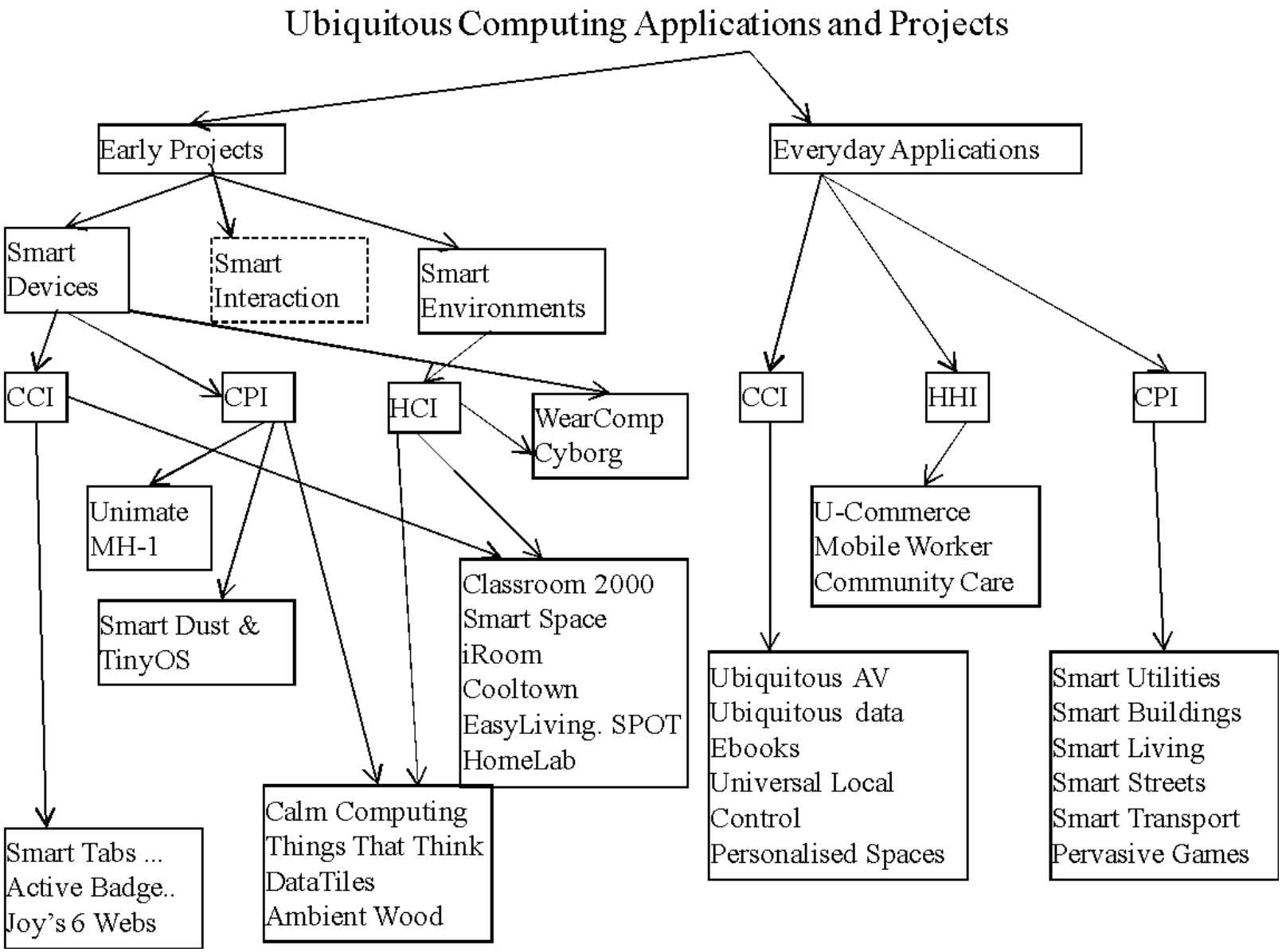
Example of Ubiquitous Computing Early Projects

DEI	Application	Properties		Interaction	
D	Smart Boards, Pads and Tabs (1990s)	Early distributed devices enabling shared computing	Distributed	Multiple devices communicate and share data	CCI
D	Active Badge, Bat and Floor (1992)	Mobility, Location awareness	Context Awareness	Tracks user position via sensors	CCI
E	Classroom 2000 (1999)	Record lectures	Context Awareness	Records and processes classroom activities	CCI CPI
E	Smart Space and Meeting Room (2003)	Taking notes	Context Awareness & Distributed	LAN support Wireless button	CCI CPI
E	Interactive Workspaces and iRoom (2002)	Assist speaker, integrated devices and displays for interaction	Context Awareness & Distributed	Devices coordinate in workspace	CCI CPI
E	Cooltown (2003)	Links physical spaces to web services	Context Awareness	Location-based access to services	CCI CPI
E	EasyLiving (Microsoft 2002)	Services based on user presence and context	Context Awareness & iHCI	Sensors + devices adapt to users	CCI CPI
E	SPOT (Microsoft 2003)	Smart devices delivering personal information	iHCI	Simple user interaction via devices with bluetooth/radio	CCI CPI

Example of Ubiquitous Computing Early Projects (cont.)

DEI	Application	Properties		Interaction	
E	HomeLab & Ambient Intelligence (Philips 2003, 2004)	Intelligent environments adapting to users	Context Awareness & iHCI	Environment responds to user behavior	CCI CPI
D	Unimate and MH-1 Robots	Industrial robots interacting with physical world	Distributed	Machines act on physical environment	CPI
D	Smart Dust and TinyOS (2001)	Tiny sensor networks collecting data	Distributed & Context Awareness	Sensors communicate environmental data	CPI
D	Calm Computing (1997)	Technology operates in background	iHCI	Minimal user attention required	HPI
D	Things That Think and Tangible Bits (1995)	Physical objects used for interaction	iHCI	Tangible interaction with digital data	HPI
D	DataTiles (Sony)	Interactive tiles for physical interaction	iHCI	Touch-based physical interaction	HPI
D	AmbientWood (2004)	Interactive learning environment	Context Awareness & iHCI	Sensors + user interaction in environment	HPI
D	WearComp & WearCam	Wearable devices enhancing perception	iHCI	Continuous interaction with user	HPI
D	Cyborg 1.0 & 2.0	Human-technology integration (embed)	iHCI	Direct interaction with human body	HPI

Framework: Interaction



Applications in the Virtual, Human and Physical World

CCI

- GPS
- Cellular Wireless Technology

HCI

- Ubiquitous Audio Video Content Access
- Ubiquitous Information Access
- Ebooks
- Universal Local Control of ICT Systems
- Personalized Spaces

HHI

- M Commerce, U Commerce
- Mobile Worker
- Community Care

HPI and CPI

- Smart Utilities
- Smart Buildings and Home Automation
- Smart Living Environments
- Smart Street
- Smart Vehicles, Transport and Travel
- Pervasive Games

CCI: Computer-computer interaction **CPI**: Computer-physical interaction **HHI**: Human-human interaction
HCI: Human-computer interaction **HPI**: Human-physical interaction

Applications in the Virtual, Human and Physical World

Application Area	Example Application	Sensor Type	Core Objective
Smart Healthcare	Wearable health monitoring	Heart rate sensors, body temperature sensors, motion sensors, wearable biosensors	Continuously monitor patient health and detect medical conditions in real time
Smart Homes	Smart lighting	Light sensors, motion sensors, occupancy sensors	Automatically adjust lighting based on presence and environment
	Energy management	Smart meters, temperature sensors, power consumption sensors	Optimize energy use and reduce electricity consumption
Smart Cities	Traffic monitoring	Cameras, vehicle detection sensors, GPS sensors	Monitor traffic flow and improve transportation efficiency
	Environmental monitoring	Air quality sensors, humidity sensors, temperature sensors	Monitor environmental conditions and support sustainable city management
Smart Education	Context-aware learning environments	Location sensors, user activity sensors, device interaction sensors	Adapt learning content based on student context and activity

Comparison of Applications of Ubiquitous Computing



Feature	Ubiquitous Computing (UbiCom)	Conventional (Non-Ubiquitous) Computing
User-to-Device Ratio	**Many-to-many:** One person owns/uses many computers; many people may share one environment device.	**One-to-one:** One person dedicated to one personal computer.
Physical Form	**Diverse & Embedded:** Heterogeneous devices (dust, tabs, pads, boards) embedded in furniture, clothes, and cars.	**Macro & Standardized:** Desktops or laptops with fixed monitors, keyboards, and mice.
Interaction Focus	**Human-centric:** Technology "disappears" or "gets out of the way" to support the task at hand.	**Computer-centric:** User must sit, stare at a screen, and "tinker" with the system to complete tasks.
Network & Connectivity	**Ad hoc & Volatile:** Spontaneous, wireless connections that change frequently as devices move.	**Fixed & Stable:** Relies on wired LANs or stable wireless with fixed network addresses.
Context Awareness	**Context-aware:** Senses and adapts to location, time, user preferences, and environmental conditions.	**Context-blind:** Primarily acts as a "dumb" terminal for manual data entry.
Goal of Technology	**Embodied Virtually:** Pushes digital functionality back into the physical world (e.g., a smart book).	**Virtualization:** Tries to virtualize the world (e.g., the "desktop" icon).

Conclusion

Key points:

- Ubiquitous computing are implemented in various **applications.**

- **Issue & Challenges** in **Ubiquitous Computing?**

